

CAESAR CIPHER WHEEL

A Simple Application of Arithmetic Modulo

Mathematics Project

AMAL JYOTHI COLLEGE OF ENGINEERING (AUTONOMOUS)

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1. ABSTRACT

The Caesar Cipher Wheel is a simple application of Arithmetic Modulo used to demonstrate the basic concept of encryption and decryption. A Caesar Cipher works by shifting each letter of a message by a fixed number of positions in the alphabet, where the fixed number is called the shift or key. Since the English alphabet contains 26 letters, modulo 26 is used to handle the wrap-around of letters.

This project demonstrates the concept using a physical cardboard Caesar Cipher Wheel and an interactive web application. The physical model provides a visual representation of the shifting process, while the website demonstrates the same concept digitally.

The project connects the mathematical concept of Arithmetic Modulo with a simple practical application in cryptography.

2. INTRODUCTION

The Caesar Cipher Wheel is a simple and basic form of encryption based on Arithmetic Modulo.

A Caesar Cipher works by shifting each letter of a message by a fixed number of positions in the alphabet. This fixed number is called the shift or key.

This project demonstrates the concept using:

- A physical cardboard Caesar Cipher Wheel
- An interactive web application

The project focuses on showing how a mathematical concept, Arithmetic Modulo, can be applied to encryption and decryption.

3. OBJECTIVES

The objectives of this project are:

1. To understand the basic concept of the Caesar Cipher.
2. To demonstrate the application of Arithmetic Modulo in encryption and decryption.
3. To understand how a fixed shift can transform plaintext into ciphertext.
4. To understand how modulo 26 provides alphabet wrap-around.
5. To demonstrate the concept using both a physical model and an interactive website.

4. BASIC CONCEPT OF CAESAR CIPHER

A Caesar Cipher is a simple and basic form of encryption. It works by shifting each letter by a fixed number of positions in the alphabet. The fixed number is called the shift or key.

4.1 Example with a Shift of 3

With a shift of 3, each letter is replaced by the letter three positions after it:

Plaintext Letter	Ciphertext Letter
A	D
B	E
C	F

4.2 Wrap-Around

When the end of the alphabet is reached, the letters wrap around to the beginning. For example, with a shift of 3:

Plaintext Letter	Ciphertext Letter
X	A
Y	B
Z	C

4.3 Convention Used in Our Implementation

Convention: In our implementation, we use a forward/right shift for encryption and the corresponding reverse/left shift for decryption. This is the convention followed throughout this report, and it is the one used in both the physical wheel and the website.

This follows the mathematical formulation:

Encryption:

$$C = (P + K) \bmod 26$$

Decryption:

$$P = (C - K) \bmod 26$$

5. CAESAR CIPHER WHEEL

Our physical Caesar Cipher Wheel consists of two alphabet circles:

- **Outer circle** — fixed
- **Inner circle** — rotates according to the selected shift

The inner wheel is rotated according to the number of shifts. For example, with a shift of 3, the inner circle is shifted by three positions. The corresponding letters on the two circles give the encrypted or decrypted letters. The wheel can be used to demonstrate both encryption and decryption.

For our implementation:

Operation	Direction of Shift
Encryption	Forward / right shift
Decryption	Reverse / left shift



Figure 1: Cardboard Caesar Cipher Wheel

6. ARITHMETIC MODULO

Arithmetic Modulo is the main mathematical concept of this project. There are 26 letters in the English alphabet, so we use modulo 26.

6.1 Alphabet Numbering

The letters are represented using numbers from 0 to 25:

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Table: Alphabet numbering ($A = 0, B = 1, \dots, Z = 25$)

6.2 Meaning of “mod”

The expression “ $a \bmod n$ ” means the remainder obtained when a is divided by n . Modulo 26 therefore allows the alphabet to wrap around whenever a calculation goes beyond Z.

6.3 Example: $Z + 3$

$Z = 25$, so with a shift of 3:

$$25 + 3 = 28$$

$$28 \bmod 26 = 2$$

Since $28 = 1 \times 26 + 2$, the remainder is 2. The number 2 represents C. Therefore $Z \rightarrow C$.

6.4 Example: A Value Smaller Than 26

$$5 \bmod 26 = 5$$

This is because 5 divided by 26 gives a quotient of 0 and a remainder of 5. Hence, a value smaller than 26 is unchanged by modulo 26, which reinforces the meaning of the modulo operation.

7. MATHEMATICAL FORMULATION

7.1 Encryption

$$C = (P + K) \bmod 26$$

Where:

- P = numerical position of the plaintext letter
- K = shift / key
- C = numerical position of the ciphertext letter

The addition of K represents the forward/right shift used in our implementation.

Example: Encrypt C using a shift of 3.

$C = 2$, so:

$$C = (2 + 3) \bmod 26 = 5 \bmod 26 = 5$$

The number 5 represents F. Therefore $C \rightarrow F$.

7.2 Decryption

$$P = (C - K) \bmod 26$$

Where:

- C = numerical position of the ciphertext letter
- K = shift / key
- P = numerical position of the plaintext letter

The subtraction of K represents the reverse/left shift.

Example: Decrypt F using a shift of 3.

$F = 5$, so:

$$P = (5 - 3) \bmod 26 = 2$$

The number 2 represents C. Therefore $F \rightarrow C$.

8. COMPLETE WORKED EXAMPLE

Word: MATHS **Shift / key:** 3

8.1 Encryption

Plaintext	Value	Shift	Ciphertext
M	12	+3	P
A	0	+3	D
T	19	+3	W
H	7	+3	K
S	18	+3	V

Calculation using $C = (P + K) \bmod 26$:

Letter	Calculation	Result
M	$(12 + 3) \bmod 26 = 15$	P
A	$(0 + 3) \bmod 26 = 3$	D
T	$(19 + 3) \bmod 26 = 22$	W
H	$(7 + 3) \bmod 26 = 10$	K
S	$(18 + 3) \bmod 26 = 21$	V

Therefore: MATHS → PDWKV

8.2 Decryption

Decryption reverses the shift, using the same key of 3.

Ciphertext	Value	Shift	Plaintext
P	15	-3	M
D	3	-3	A
W	22	-3	T
K	10	-3	H
V	21	-3	S

Calculation using $P = (C - K) \bmod 26$:

Letter	Calculation	Result
P	$(15 - 3) \bmod 26 = 12$	M
D	$(3 - 3) \bmod 26 = 0$	A
W	$(22 - 3) \bmod 26 = 19$	T
K	$(10 - 3) \bmod 26 = 7$	H
V	$(21 - 3) \bmod 26 = 18$	S

Therefore: PDWKV → MATHS

9. DIGITAL IMPLEMENTATION

Our project also includes an interactive HTML/web application. The website demonstrates the same Caesar Cipher process digitally. The website allows the user to:

- Enter text
- Select encryption or decryption
- Select a shift/key
- Perform encryption
- Perform decryption
- View the resulting text
- Visualize the shifting process

9.1 Demonstration

Input	Operation	Shift	Result
MATHS	Encrypt	3	MATHS → PDWKV
PDWKV	Decrypt	3	PDWKV → MATHS

The website follows the same mathematical principles as the physical wheel.

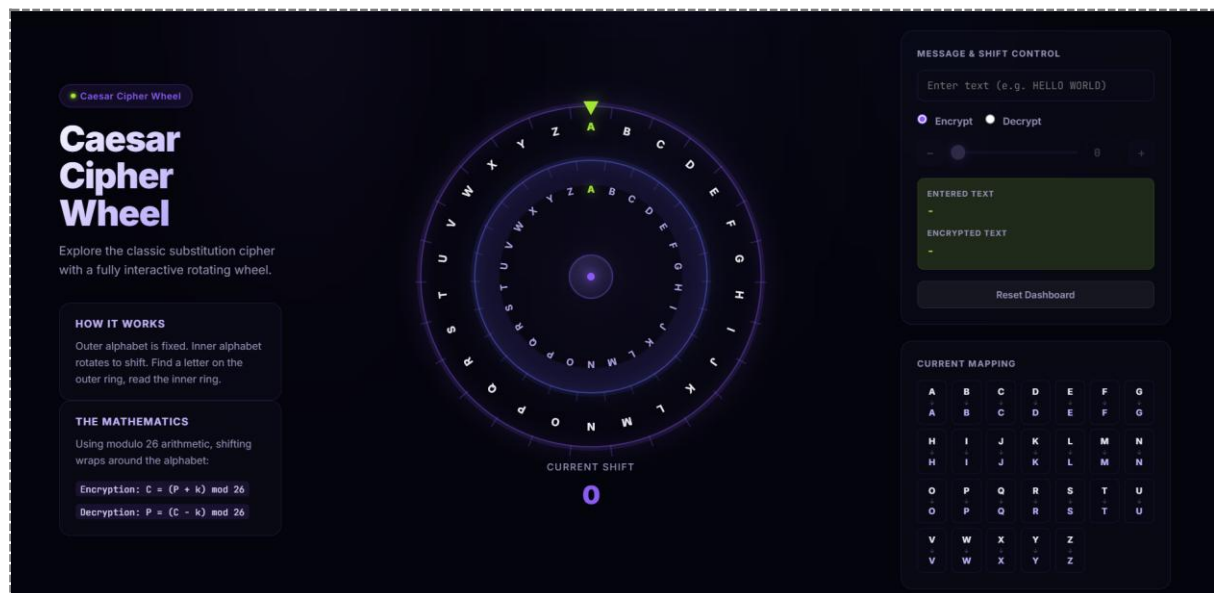


Figure 2: Caesar Cipher Wheel website

10. LIMITATIONS

Although the Caesar Cipher is useful for demonstrating encryption and Arithmetic Modulo, it has several limitations:

- There are only 26 possible shift values.
- All possible shifts can easily be tested.
- It can therefore be broken using a simple brute-force approach.
- It is not suitable for modern secure communication.

However, it is useful for learning the fundamentals of:

- Encryption
- Decryption
- Modular arithmetic
- Alphabet shifting
- Basic cryptographic concepts

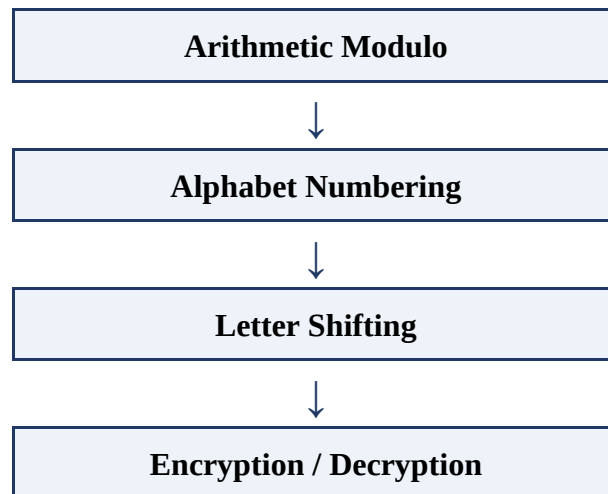
11. CONCLUSION

The Caesar Cipher Wheel provides a simple way to understand how Arithmetic Modulo can be applied to encryption and decryption.

The physical cardboard model provides a visual representation of alphabet shifting, while the interactive website demonstrates the same process digitally.

The use of modulo 26 allows the alphabet to wrap around when a shift goes beyond Z. The mathematical formulas provide a precise method for performing encryption and decryption.

The project therefore connects the mathematical concept of Arithmetic Modulo with a practical application of encryption. The key relationship demonstrated by the project is:



12. REFERENCES

The sources used for this project are listed below.

- [1] QuickCS, “Caesar Cipher Encryption and Decryption Example,” YouTube, 2017.
<https://www.youtube.com/watch?v=k8W2SgaIkSY>
Used for: understanding the basic Caesar Cipher, shifting, encryption and decryption.

- [2] MDN Web Docs, “String – JavaScript,” Mozilla Developer Network.
https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/String
Used for: website/JavaScript implementation reference.

- [3] Princeton University, “Caesar’s Cipher: A Closer Look,” COS 116 lecture notes.
<https://www.cs.princeton.edu/courses/archive/spr07/cos116/lectures/lec19.pdf>
Used for: mathematical explanation of the Caesar Cipher and modulo 26.

13. INDIVIDUAL CONTRIBUTION

Name	KRISHNAJITH H
Role	Website Development and Presentation

Contribution

I independently designed and developed the interactive Caesar Cipher Wheel website used in the project. I implemented and tested the encryption and decryption functionality, shift-based alphabet transformation, and user interface. I verified the results using different shift values.

I also participated in the presentation and explained the mathematical concepts behind the Caesar Cipher, including alphabet numbering, modulo 26, and the encryption and decryption formulas.